DATA CLASSIFICATION AND FEATURE ENCODING

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TASK 1 – QUANTITATIVE VARIABLES

1. Rectal Temperature (rectal_temp)
   level: interval – measured in °C
   state: continuous – can take any fractional value
2. Total Protein (total_protein)
   level: ratio – has a true absolute zero
   state: continuous – measured to decimal fractions
3. Pulse (pulse)
   level: ratio – 0 bpm = no heartbeat
   state: discrete – heartbeats counted as indivisible events

TASK 1 – QUALITATIVE VARIABLES

1. Clinical Outcome (outcome)
   level: nominal – no mathematical rank
   state: discrete
2. Pain level (pain)
   level: ordinal – clear progression
   state: discrete
3. Surgery Performed (surgery)
   level: nominal – yes/no
   state: discrete

TASK 2 – ML PREPARATION

- Feature selected: Clinical Outcome (outcome)
- Technique chosen: dummy encoding
- Challenge: encode “lived“, “died“ and “euthanized“.
- Using ordinal encoding (1, 2, 3) the model would incorrectly assume lived (1) is smaller than euthanized (3)

TASK 2 – DUMMY ENCODING STEPS

1. Identify categories
2. Generate binary matrices
3. Assign boolean values
4. Avoid the dummy variable trap

CONCLUSION

Summary:

- Accurate data classification (Nominal vs. Ordinal, Interval vs. Ratio) dictates the statistical tools available.
- Feature encoding bridges the gap between clinical/qualitative lab data and quantitative algorithms.

Application:

- These prepared variables are now ready for predictive modeling (e.g., Logistic Regression or Random Forests) to forecast colic survival rates.